

# Low-frequency and broadband vibration self-suppression of fluid-conveying pipelines via metamaterial-inspired design

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## ARTICLE INFO

### Keywords:

Pipeline meta structure

Metamaterial

Vibration suppression

Acoustic black hole

Band Gaps

## ABSTRACT

Phononic crystal (PC) pipelines conveying fluid have been widely utilized in engineering structures for vibration suppression. However, their practical application remains constrained by two persistent challenges: the difficulty in achieving simultaneous low-frequency and broadband vibration attenuation, and the lack of effective strategies for directional band gap (BG) design. Inspired by metamaterial concepts, this study presents a hierarchical pipeline meta structure composed of periodically arranged lattices. Each lattice incorporates a pair of oppositely oriented sub-lattices with acoustic black hole (ABH) geometries. Through mechanical modeling based on Timoshenko beam theory and semi-analytical solutions combining the spectral element method (SEM) with the transfer matrix method (TMM), the dynamic characteristics including natural frequencies, frequency responses, and BGs are systematically investigated. The results demonstrate that the band gap (BG) width not only determines the vibration suppression frequency range but also exhibits a positive correlation with attenuation effectiveness. An increase in the flow velocity of the conveyed fluid adversely affects the stability of the pipeline meta structure. However, within the critical flow velocity range, its impact on the BGs remains negligible. Furthermore, BG characteristics are simultaneously governed by sublattice configuration and ABH geometric parameters. Specifically, asymmetric lattice designs incorporating different materials or geometric configurations facilitate broadband BG formation. The ABH parameters offer precise tunability: increasing the power-law exponent induces a low-frequency shift of the BGs, whereas enlarging the maximum outer diameter effectively broadens the BGs' bandwidth. These findings present a promising strategy for overcoming current limitations in periodic pipeline applications.

## 1. Introduction

Fluid-conveying pipelines, as a prevalent type of transportation equipment, are extensively integrated into a wide range of engineering systems, such as hydraulic control systems, gas transmission networks, offshore drilling platforms, and nuclear reactor cooling systems. However, pipeline vibrations pose a significant threat to both the structural and functional integrity of engineering

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infrastructures. The primary mechanisms driving these risks include vibration-induced mechanical degradation, such as wear and fatigue processes, as well as the concurrent generation of noise pollution. From one perspective, environmental excitation propagates along the pipeline through its support structures and radiates into the main structural system, effectively transforming the pipeline into a medium for wave propagation. From the other perspective, the flow-induced vibration caused by the conveyed fluid establishes the pipeline itself as an inherent vibration source. These two interrelated mechanisms collectively constitute a significant challenge for achieving effective vibration control in fluid-conveying pipelines.

In recent literature, the primary approaches to vibration control in fluid-conveying pipelines have been categorized into passive control, active control, and structural optimization design. Among these, passive control primarily relies on incorporating structural damping into the pipeline's main structure. This method is straightforward to implement and has been successfully applied in various engineering systems, including railway tracks [1], bridges [2], ship hulls [3], and automotive suspensions [4]. However, conventional structural damping exhibits limited effectiveness in low-frequency vibration suppression. For fluid-conveying pipelines, low-frequency vibration control is particularly critical due to their large axial dimensions and inherent fluid-structure interaction characteristics. Advanced vibration mitigation strategies, such as nonlinear energy sink (NES)-based absorbers [5], quasi-zero stiffness (QZS) local structures [6], and friction- or impact-based energy dissipation devices [7–10], have partially overcome the low-frequency limitations of traditional passive techniques. Nevertheless, these designs suffer from narrow attenuation bandwidths, making it challenging to achieve simultaneous low-frequency and broadband vibration control. Active control methods have achieved significant theoretical success owing to their effective vibration suppression capabilities across both low- and high-frequency ranges. In fluid-conveying pipeline systems, active control architectures typically consist of an energy supply, controllers, sensors, and actuators, where the latter are commonly implemented as piezoelectric or electromagnetic structures [11]. Notable examples include Chen et al. [12], who demonstrated effective vibration control of cantilevered pipelines using hard-magnetic soft material-based actuators, and Liang and Chen [13], who enhanced bending stability and achieved systematic vibration suppression in fluid-conveying pipelines through a smart piezoelectric feedback structure. However, practical engineering applications of active control remain extremely limited due to the complexity of system architecture and the continuous requirement for external energy input. In certain fluid-conveying pipeline systems, such as aircraft hydraulic control lines [14] and drill strings [15], the demanding operational environments impose severe constraints, which hinder the reliable and sustained performance of active control systems.

Compared to active and passive control methods, structural optimization of pipelines, such as rational support layout, offers a more practical and effective approach to improving their dynamic performance [16]. As an emerging paradigm in design optimization, metamaterials have been widely adopted for vibration and noise mitigation in various engineering structures. The core principle of vibration reduction using metamaterials lies in the design of periodic lattice structures that attenuate or block elastic waves within specific frequency ranges. These frequency ranges are known as band gaps (BGs), and the periodic structures capable of controlling elastic wave propagation are commonly referred to as phononic crystals (PCs). The wave control capability of PCs can be realized through periodic boundary conditions [17], topological designs, or multi-physics coupling mechanisms. For instance, Zhang and Bian successfully tuned the BG of topological PCs by leveraging a magneto-thermal coupled field [18]. In recent years, PC pipes have been proposed for elastic wave control, with applications including vibration suppression. The underlying mechanisms for bandgap generation can primarily be categorized into two types: Bragg scattering and local resonance (LR). For the Bragg scattering type, structures that have received significant attention include the binary PC pipe proposed by Yu et al. [19], the periodic three-step pipe introduced by Liang et al. [20], and the electro-mechanical meta-riser pipe developed by Liang [21]. A defining characteristic of the local resonance type is the presence of periodically arranged local equivalent resonators [22]. Notable examples of successful designs include periodic quasi-zero-stiffness ring configurations [23] and periodic composite material ring configurations [24]. In addition to the aforementioned designs based on periodic materials or geometric configurations and periodical oscillator structures, recent studies have reported various lattice-level microstructural design strategies, such as the incorporation of chirality [25], hierarchy [26], and Symmetry breaking [27] for enhancing the mechanical properties of metamaterials.

Classic PC pipes exhibit several key performance limitations. For Bragg scattering types, the position and width of the BGs are constrained by the lattice size, achieving a low-frequency wideband BG requires relatively large lattice dimensions, which poses significant challenges for practical implementation in engineering systems. In contrast, local resonance types typically exhibit narrow BGs, limiting their ability to meet the demands of real-world applications for broadband vibration suppression, resulting in suboptimal damping performance [28]. It is noteworthy that in phononic crystal (PC) pipes, the wave regulation mechanisms of different structural types are not mutually exclusive. The work by Liang et al. demonstrates that local resonance (LR) bandgaps and Bragg bandgaps can be coupled to form a hybrid BG [29]. In addition to PC structures, another widely investigated approach for wave manipulation is the acoustic black hole (ABH), which features an outer contour with a power-law profile. This design enables compression of bending wave wavelengths, reduces the phase velocity of structural waves, and concentrates wave energy into a confined region, thereby effectively suppressing wave propagation [30]. ABH structures exhibit favorable low-frequency broadband wave regulation capabilities [31,32]. By arranging ABH elements periodically or incorporating hollow configurations, such structures can be engineered as metamaterials [33,34] for applications in vibration suppression [35,36], energy harvesting [37], and signal sensing [38]. Inspired by the BG generation mechanisms in PC pipes and the inherent low-frequency broadband characteristics of the ABH effect, this study integrates these two principles to propose a novel pipeline meta structure designed for low-frequency broadband vibration mitigation. In addition, the authors incorporate a hierarchical geometric design into the lattice of the proposed pipeline meta structure to further mitigate the dependence of BG characteristics on lattice size.

The remainder of this paper is organized as follows: Section 2 presents the design of the pipeline meta structure and establishes its mechanical model. Section 3 introduces the numerical method used to solve the dynamic behavior of the model developed in Section 2. Section 4 presents the band structure characteristics and dynamic response of the proposed pipeline meta structure, with a detailed

analysis of bandgap tuning mechanisms. Section 5 summarizes the main conclusions of this study.

## 2. Modeling of periodical fluid-conveying pipes

Fig. 1 presents fluid-conveying pipelines in various engineering systems, such as marine piping systems (Fig. 1a), aircraft hydraulic control lines (Fig. 1b) and drill strings (Fig. 1c). Due to the diverse operating conditions across these applications, a universal vibration mitigation strategy based on active or passive control methods is challenging to implement. In contrast, replacing conventional pipelines with periodic configurations that exploit bandgap mechanisms offers a more broadly applicable and inherently robust approach to vibration suppression.

Fig. 2 displays the design of a metamaterial-inspired periodic fluid-conveying pipes, in which a unit cell comprises two geometrically inverted sub-lattices that together form a hierarchical architecture, labeled as part-A and part-B. These sub-lattices feature the ABH geometry, characterized by the outer diameter that follows a power-law variation, with the maximum outer diameter denoted as  $r_o$  and the minimum as  $r_c$ . The inner diameter remains constant throughout the structure and is designated as  $r_i$ . Notably, part-A and part-B can be fabricated from the same material or configured as a hybrid composite using dissimilar materials. Their respective dynamic behaviors will be systematically analyzed and presented in the results section.

By neglecting the nonlinear terms, the motion equation for a straight fluid-conveying pipe undergoing fluid-structure interaction (FSI) can be established based on Timoshenko beam theory and Hamilton's principle [40].

$$EI(\kappa GA_p - m_f U^2) w'''' + J(m_f + m_p) \ddot{w} - 2EI m_f U \dot{w}''' - \kappa GA_p m_f U^2 w'' + 2J m_f U \dot{w}' - [EI(m_f + m_p) + J(\kappa GA_p - m_f U^2)] \dot{w}' + 2\kappa GA_p m_f U \dot{w}' + \kappa GA_p (m_f + m_p) \dot{w} = 0 \quad (1)$$

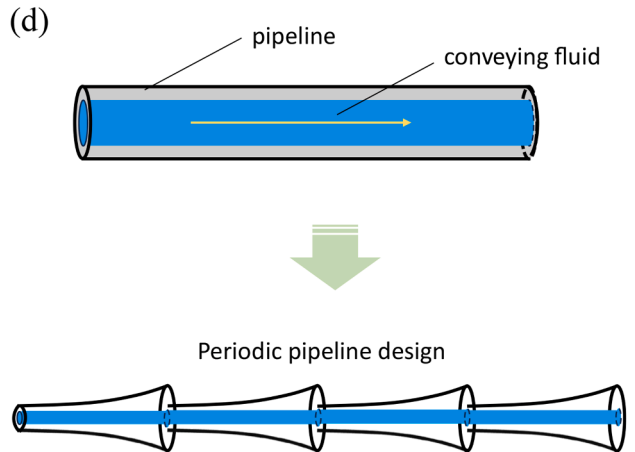
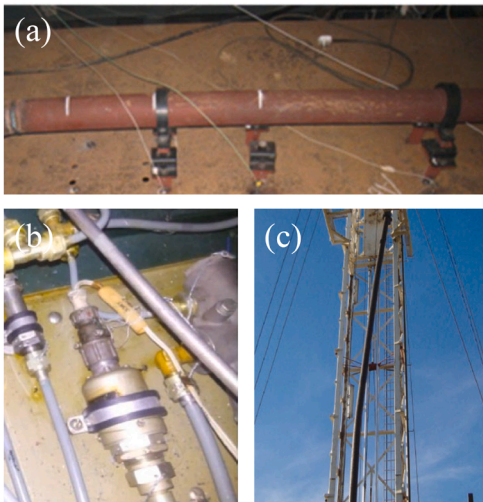
where  $(\cdot)'$  and  $(\cdot)''$  represent the partial derivatives of the variable with respect to the spatial coordinate  $x$  and time  $t$ , respectively.  $m_f$  and  $U$  denote the mass of liquid per unit length and the flow velocity of the conveyed fluid, respectively. When these parameters are set to zero, Eq. (1) reduces to the forced vibration equation of a Timoshenko beam.  $E$  and  $G$  are the Young's modulus and shear modulus of the current pipe.  $A_p$ ,  $m_p$  and  $J$  represent cross-sectional area of the pipe, the mass of pipe per unit length and the moment of inertia of the unit-length fluid-carrying pipe about the neutral axis, respectively, while  $\kappa$  is the section shear coefficient in the Timoshenko model. For a thin-walled pipe,  $\kappa = 2(1 + \mu)/(4 + 3\mu)$ . Based on the Fourier transform,  $w$  can be expressed as a superposition of spectral functions, denoted by

$$w(x, t) = \frac{1}{N_s} \sum_{n=1}^{N_s} \Phi_n(x, \omega) e^{i\omega t}, \quad (2)$$

where  $i$  is the imaginary unit and  $\omega = 2\pi f$  which represents the circular frequency where  $f$  is the incident frequency.  $N_s$  represents the number of samples in the time domain, and it is usually taken as an integer power of 2.  $\Phi$  denotes the shape functions in frequency domain. In Eq (1), it takes a harmonic form and is expressed as

$$\Phi_n(x, \omega) = \varphi_n e^{ikx}, \quad (3)$$

where  $\varphi_n$  represents the spectral amplitude and  $k$  is the wave number, which depends on the frequency  $\omega$  and is determined by solving



**Fig. 1.** Fluid-conveying pipes: (a) Pipes connected to a ship's hull for fluid transport [39], (b) Drill strings used in drilling operations [15], (c) Hydraulic tubing systems in aircraft [14], (d) PC pipeline for vibration reduction.

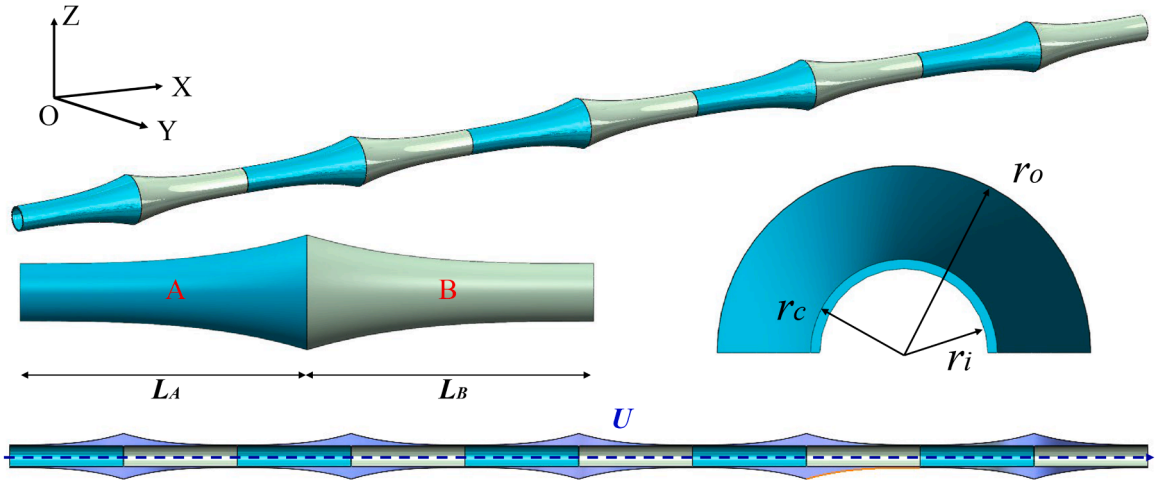


Fig. 2. Geometry of the metamaterial-inspired pipeline conveying fluid.

the dispersion equation. This equation is obtained by substituting Eqs. (2) and (3) into Eq. (1),

$$EI(\kappa GA_p - m_f U^2)k^4 - 2m_f U EI \omega k^3 - [(\kappa GA_p - m_f U^2)J + EI(m_f + m_p)]\omega^2 k^2 + \kappa GA_p m_f U^2 k^2 - 2m_f U (\kappa GA_p - J\omega^2)\omega k + (J\omega^2 - \kappa GA_p)(m_f + m_p)\omega^2 = 0 \quad (4)$$

Once the physical parameters of the flow transmission pipeline are fixed, solving Eq. (4) yields two pairs of complex conjugate roots associated with the wavenumber, which are denoted as  $k_1$ ,  $k_2$ ,  $k_3$  and  $k_4$ . Therefore, for Eq. (3), a suitable sampling rate is  $N_s = 4$ . Substituting Eq. (4) into Eqs. (2) and (3), and incorporating the relationship between generalized forces and displacements in the Timoshenko beam model, all expressions for the generalized forces and generalized displacements can be formulated as

$$\begin{cases} w(x, t) = \sum_{n=1}^4 \varphi_n e^{i(\omega t + k_n x)} \\ \theta(x, t) = \sum_{n=1}^4 \Gamma_n \varphi_n e^{i(\omega t + k_n x)} \\ Q(x, t) = \kappa GA_p \sum_{n=1}^4 (ik_n - \Gamma_n) \varphi_n e^{i(\omega t + k_n x)} \\ M(x, t) = EI \sum_{n=1}^4 ik_n \Gamma_n \varphi_n e^{i(\omega t + k_n x)} \end{cases}, \quad (5)$$

where  $\theta$  is the slope,  $Q$  and  $M$  represent the shear force and the bending moment, respectively.  $\Gamma_n$  is the coefficient determined by the

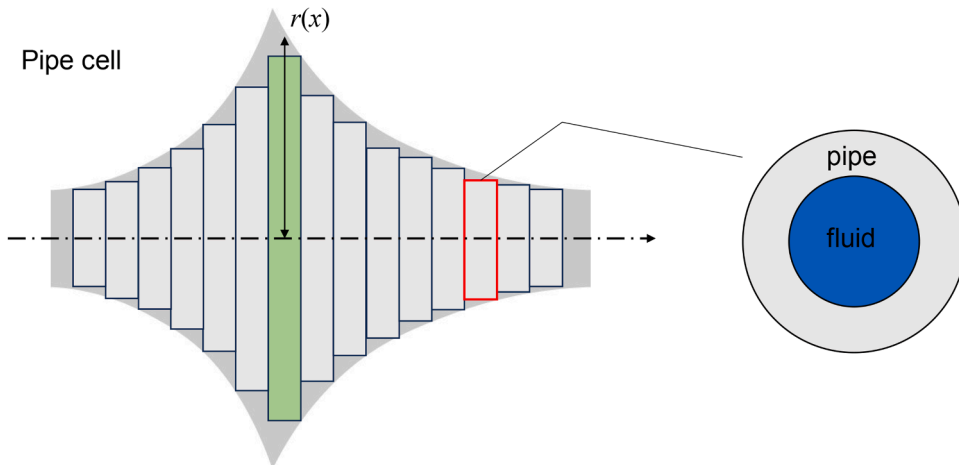


Fig. 3. Single-period discretization.

relationship of deflection and slope in the Timoshenko beam [29], as follows

$$\Gamma_n = \theta/w = i\kappa GA_p k_n / (EI k_n^2 + \kappa GA_p - J\omega^2). \quad (6)$$

### 3. Computational Methods

To investigate the wave propagation behavior in fluid-conveying pipes, this study employs the spectral element method (SEM) in conjunction with the transfer matrix method (TMM) for the numerical solution of Eqs. (1) to (6). Based on this approach, the frequency response and wave mode shapes of the pipeline meta structure are explicitly presented. Additionally, the procedures for calculating the BG characteristics and natural frequencies are developed within this section.

For a pipeline meta structure featuring a variable cross-section, discretization into multiple segments with uniform radius is required to meet the conditions of Eq. (1). Given the ABH geometry characterized by a gradual change in outer diameter, a stepped approximation using a series of segmented pipe elements is adopted to emulate the tapered profile of the ABH, as shown in Fig. 3. The outer diameter of each element is defined according to the local thickness of the ABH's outer edge at that position, computed as

$$\begin{cases} r(n) = (r_o - r_c) \left( \frac{n-1}{N_A} \right)^\beta + r_c, & \text{in part - A} \\ r(n) = (r_o - r_c) \left[ \frac{L_A - \frac{L_A}{N_A} \cdot (n - N_A - N_B - 1)}{L_A} \right]^\beta + r_c, & \text{in part - B} \end{cases} \quad (7)$$

where  $N_A$  and  $N_B$  denote the number of elements in part-A and part-B within a single unit cell, respectively, while  $\beta$  refers to the power-law exponent characterizing the ABH profile. As the element length approaches sufficiently small values, the stepped composite pipe model converges to an exact representation of the pipeline meta structure depicted in Fig. 2. Within the local coordinate, the generalized forces and displacements at the left and right terminations of any given pipe element can be expressed in terms of their corresponding local coordinate variables, formulated as

$$w_1 = w(0, t), \quad \theta_1 = \theta(0, t), \quad w_2 = w(L_e, t), \quad \theta_2 = \theta(L_e, t), \quad (8)$$

and

$$Q_1 = -Q(0, t), \quad M_1 = -M(0, t), \quad Q_2 = Q(L_e, t), \quad M_2 = M(L_e, t), \quad (9)$$

where  $Q_1$  and  $Q_2$  denote the variables at the left and right ends, respectively, and  $L_e$  signifies the length of the current element. By separately substituting Eqs. (8) and (9) into Eq. (5), the transfer relationships between the generalized forces and generalized displacements at the two ends of the element can be derived as follows:

$$\mathbf{D}_e = \mathbf{L} \varphi e^{i\omega t}, \quad (10)$$

where

$$\mathbf{L} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ \Gamma_1 & \Gamma_2 & \Gamma_3 & \Gamma_4 \\ e^{iL_e k_1} & e^{iL_e k_2} & e^{iL_e k_3} & e^{iL_e k_4} \\ \Gamma_1 e^{iL_e k_1} & \Gamma_2 e^{iL_e k_2} & \Gamma_3 e^{iL_e k_3} & \Gamma_4 e^{iL_e k_4} \end{bmatrix}, \quad (11)$$

and

$$\mathbf{F}_e = \mathbf{R} \varphi e^{i\omega t}, \quad (12)$$

where

$$\mathbf{R} = \begin{bmatrix} \kappa GA_p (ik_1 - \Gamma_1) & \kappa GA_p (ik_2 - \Gamma_2) & \kappa GA_p (ik_3 - \Gamma_3) & \kappa GA_p (ik_4 - \Gamma_4) \\ iEI k_1 \Gamma_1 & iEI k_2 \Gamma_2 & iEI k_3 \Gamma_3 & iEI k_4 \Gamma_4 \\ -\kappa GA_p (ik_1 - \Gamma_1) e^{iL_e k_1} & -\kappa GA_p (ik_2 - \Gamma_2) e^{iL_e k_2} & -\kappa GA_p (ik_3 - \Gamma_3) e^{iL_e k_3} & -\kappa GA_p (ik_4 - \Gamma_4) e^{iL_e k_4} \\ -iEI k_1 \Gamma_1 e^{iL_e k_1} & -iEI k_2 \Gamma_2 e^{iL_e k_2} & -iEI k_3 \Gamma_3 e^{iL_e k_3} & -iEI k_4 \Gamma_4 e^{iL_e k_4} \end{bmatrix} \quad (13)$$

For convenience, let  $\mathbf{D}_e = [\mathbf{d}_n, \mathbf{d}_{n+1}]^T = [v_n, \theta_n, v_{n+1}, \theta_{n+1}]^T$  denote the displacement vector of the  $n$ -th element, and  $\mathbf{F}_e = [\mathbf{f}_n, \mathbf{f}_{n+1}]^T = [Q_n, M_n, Q_{n+1}, M_{n+1}]^T$  represents the corresponding force vector. By integrating Eqs. (10) and (12) and removing the time-dependent component, the following expression is obtained

$$\mathbf{D}_e = \mathbf{S}_{e,n} \mathbf{F}_e, \quad (14)$$

where  $\mathbf{S}_{e,n}$  denotes the spectral stiffness matrix associated with the  $n$ -th element within the lattice, it follows

$$\mathbf{S}_{e, n} = \mathbf{R}\mathbf{L}^{-1}. \quad (15)$$

A lattice consisting of  $N$  elements features  $N + 1$  nodes, with the  $n$ -th element connected to nodes numbered  $n$  and  $n+1$ . When focusing solely on the left and right ends of the  $n$ -th element, Eq. (14) can be reformulated as

$$\begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_{n+1} \end{bmatrix} = \begin{bmatrix} \mathbf{p}_{n,11} & \mathbf{p}_{n,12} \\ \mathbf{p}_{n,21} & \mathbf{p}_{n,22} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_{n+1} \end{bmatrix}, \quad (16)$$

where

$$\begin{bmatrix} \mathbf{p}_{n,11} & \mathbf{p}_{n,12} \\ \mathbf{p}_{n,21} & \mathbf{p}_{n,22} \end{bmatrix} = \mathbf{S}_{e, n}. \quad (17)$$

By applying Eq. (16) to the first and second elements respectively, a system of coupled equations can be established.

$$\begin{cases} \mathbf{f}_1 = \mathbf{p}_{1,11}\mathbf{u}_1 + \mathbf{p}_{1,12}\mathbf{u}_2 \\ \mathbf{f}_2 = \mathbf{p}_{1,21}\mathbf{u}_1 + \mathbf{p}_{1,22}\mathbf{u}_2 \\ \mathbf{f}_2 = \mathbf{p}_{2,21}\mathbf{u}_2 + \mathbf{p}_{2,22}\mathbf{u}_3 \\ \mathbf{f}_3 = \mathbf{p}_{2,21}\mathbf{u}_2 + \mathbf{p}_{2,22}\mathbf{u}_3 \end{cases}. \quad (18)$$

By solving and eliminating  $\mathbf{f}_2$  and  $\mathbf{u}_2$  from Eq. (18), one obtains

$$\begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{T}_{2,11} & \mathbf{T}_{2,12} \\ \mathbf{T}_{2,21} & \mathbf{T}_{2,22} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_3 \end{bmatrix}. \quad (19)$$

where

$$\begin{cases} \mathbf{T}_{2,11} = \mathbf{p}_{1,11} - \mathbf{p}_{1,12}(\mathbf{p}_{1,22} + \mathbf{p}_{1,11})^{-1}\mathbf{p}_{1,21} \\ \mathbf{T}_{2,12} = -\mathbf{p}_{1,12}(\mathbf{p}_{1,22} + \mathbf{p}_{1,11})^{-1}\mathbf{p}_{1,12} \\ \mathbf{T}_{2,21} = -\mathbf{p}_{1,21}(\mathbf{p}_{1,22} + \mathbf{p}_{1,11})^{-1}\mathbf{p}_{1,21} \\ \mathbf{T}_{2,22} = \mathbf{p}_{1,22} - \mathbf{p}_{1,21}(\mathbf{p}_{1,22} + \mathbf{p}_{1,11})^{-1}\mathbf{p}_{1,12} \end{cases}. \quad (20)$$

When only the nodes at the left and right boundaries of the entire lattice are considered, with all intermediate nodes neglected, the corresponding spectral relationship should be expressed as

$$\begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_{N+1} \end{bmatrix} = \begin{bmatrix} \mathbf{T}_{N,11} & \mathbf{T}_{N,12} \\ \mathbf{T}_{N,21} & \mathbf{T}_{N,22} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_{N+1} \end{bmatrix}, \quad (21)$$

It should be noted that in Eq. (18), the submatrix  $\mathbf{T}$ , which determines the spectral stiffness matrix, must be obtained through iterative computation based on Eq. (20) [41,42], and the corresponding iteration formula is as follows:

$$\begin{cases} \mathbf{T}_{n,11} = \mathbf{p}_{n,11} - \mathbf{p}_{n,12}(\mathbf{p}_{n,22} + \mathbf{T}_{n-1,11})^{-1}\mathbf{p}_{n,21} \\ \mathbf{T}_{n,12} = -\mathbf{p}_{n,12}(\mathbf{p}_{n,22} + \mathbf{T}_{n-1,11})^{-1}\mathbf{T}_{n,12} \\ \mathbf{T}_{n,21} = -\mathbf{T}_{n-1,21}(\mathbf{p}_{n,22} + \mathbf{T}_{n-1,11})^{-1}\mathbf{p}_{n,21} \\ \mathbf{T}_{n,22} = \mathbf{T}_{n-1,22} - \mathbf{T}_{n-1,21}(\mathbf{p}_{n,22} + \mathbf{T}_{n-1,11})^{-1}\mathbf{T}_{n,22} \end{cases}. \quad (22)$$

To obtain the spectral stiffness matrix in Eq. (21), the left-hand side of Eq. (22) must be iteratively computed over the index range from  $n = 1$  to  $n = N$ . Due to the periodic arrangement of the pipe lattice, wave propagation in each unit cell must satisfy the Bloch theorem, as

$$\begin{cases} \mathbf{d}_{N+1} = e^{i\mathbf{L}q}\mathbf{d}_1 \\ \mathbf{f}_{N+1} = -e^{i\mathbf{L}q}\mathbf{f}_1 \end{cases}. \quad (23)$$

In Eq. (23),  $q$  denotes the complex wave vector, the presence of its real part corresponds to the wave's pass band, whereas the presence of its imaginary part signifies wave attenuation. Substituting Eq. (21) into Eq. (23) yields

$$[\mathbf{T}_{N,12}e^{2i\mathbf{L}q} + (\mathbf{T}_{N,11} + \mathbf{T}_{N,22})e^{i\mathbf{L}q} + \mathbf{T}_{N,21}] \mathbf{d}_1 = 0 \quad (24)$$

A nontrivial solution for Eq. (24) requires

$$\det[\mathbf{T}_{N,12}e^{2i\mathbf{L}q} + (\mathbf{T}_{N,11} + \mathbf{T}_{N,22})e^{i\mathbf{L}q} + \mathbf{T}_{N,21}] = 0 \quad (25)$$

The band structure of the fluid-conveying pipeline meta structure can be determined by solving for  $q$  in Eq. (25). For a pipeline composed of multiple lattice (with a total number of  $R$ ), the global force-displacement spectral relationship can be systematically constructed by assembling the relation given in Eq. (21), which characterizes the interactions at the left and right boundaries of a single lattice.

$$\mathbf{F}_g = \mathbf{S}_g\mathbf{D}_g, \quad (26)$$

where  $\mathbf{F}_g = [\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots, \mathbf{F}_{R+1}]$  and  $\mathbf{D}_g = [\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_3, \dots, \mathbf{D}_{R+1}]$  denote the external forces and response displacements at the interfaces between lattices, respectively. In Eq. (26), the global spectral matrix of the pipeline meta structure follows

$$\mathbf{S}_g = \begin{bmatrix} \mathbf{T}_{N,11} & \mathbf{T}_{N,12} & & & \\ \mathbf{T}_{N,21} & \mathbf{T}_{N,11} + \mathbf{T}_{N,22} & \mathbf{T}_{N,12} & & 0 \\ & \mathbf{T}_{N,21} & \mathbf{T}_{N,11} + \mathbf{T}_{N,22} & \ddots & \\ & 0 & & \mathbf{T}_{N,11} + \mathbf{T}_{N,22} & \mathbf{T}_{N,12} \\ & & & \mathbf{T}_{N,21} & \mathbf{T}_{N,22} \end{bmatrix}. \quad (27)$$

It should be noted that Eq. (26) represents the formulation for computing the system's frequency response by SEM. At the system's natural frequencies, Eq. (26) admits no non-zero solutions, and the spectral stiffness matrix must satisfy

$$\det(\mathbf{S}_g) = 0. \quad (28)$$

Furthermore, when rigid or flexible constraints are applied to the system, its natural frequencies are altered. In such cases,  $\mathbf{S}_g$  must be modified accordingly, detailed correction methods can be found in the reference [43].

In this study, the TMM is employed to validate the accuracy of the SEM results. Based on Eqs. (5), (8), and (9), the coefficient vectors  $\boldsymbol{\varphi}$  at the two nodal ends of the element exhibit the following relationship,

$$\mathbf{H}^{n,n} \boldsymbol{\varphi}^n = \mathbf{H}^{n+1,n} \boldsymbol{\varphi}^{n+1}. \quad (29)$$

where the matrix  $\mathbf{H}^{i,j}$  represents the transfer matrix of the  $j$ -th element at the  $i$ -th node, it follows

$$\mathbf{H}^{i,j} = \begin{bmatrix} e^{ik_1 x_i} & e^{ik_2 x_i} & e^{ik_3 x_i} & e^{ik_4 x_i} \\ \Gamma_1 e^{ik_1 x_i} & \Gamma_2 e^{ik_2 x_i} & \Gamma_3 e^{ik_3 x_i} & \Gamma_4 e^{ik_4 x_i} \\ \kappa G A_p (ik_1 - \Gamma_1) e^{ik_1 x_i} & \kappa G A_p (ik_2 - \Gamma_2) e^{ik_2 x_i} & \kappa G A_p (ik_3 - \Gamma_3) e^{ik_3 x_i} & \kappa G A_p (ik_4 - \Gamma_4) e^{ik_4 x_i} \\ EH \Gamma_1 k_1 e^{ik_1 x_i} & EH \Gamma_2 k_2 e^{ik_2 x_i} & EH \Gamma_3 k_3 e^{ik_3 x_i} & EH \Gamma_4 k_4 e^{ik_4 x_i} \end{bmatrix} \quad (30)$$

In Eq. (30),  $x_i$  is determined by the coordinates of the node, while the physical parameters and wavenumber are dependent on the current pipe element. Based on Eq. (29), the transfer matrix between the first node and the last node of one lattice can be obtained by [20]

$$\mathbf{Tr} = \left[ \prod_{n=N-1}^1 (\mathbf{H}^{n+1,n})^{-1} \mathbf{H}^{n,n} \right] (\mathbf{H}^{1,1})^{-1} \mathbf{H}^{N,N+1}. \quad (31)$$

By applying Bloch's theorem once again, one can obtain

$$\det(\mathbf{Tr} - e^{iL\mu} \mathbf{I}) = 0. \quad (32)$$

where the symbol  $\mathbf{I}$  denotes the identity diagonal matrix, and the BG characteristics can be quantified by Eq. (32).

## 4. Results and discussing

### 4.1. Model validation

In this section, the semi-analytical results obtained using the SEM, including wave vectors, mode shapes, and frequency responses, are presented and discussed. To validate the accuracy of the SEM results, the wave vectors are verified against results obtained via the TMM. All computations are performed on a system equipped with an AMD Ryzen 5 5800X processor and 64 GB of DDR4 RAM operating at 3200 MHz. Table 1 lists the default system parameters used in the following examples. For simplicity, these default parameters will be applied unless otherwise stated.

Fig. 4 displays the verification results of the mechanical model and the convergence behavior of the solution method. The wave

**Table 1**  
Preset system parameters.

| Symbol         | Value                  | Symbol | Value  |
|----------------|------------------------|--------|--------|
| $U$            | 0 m/s                  | $r_o$  | 0.1m   |
| $\rho_{fluid}$ | 1000 kg/m <sup>3</sup> | $r_c$  | 0.05m  |
| $\rho_{epoxy}$ | 1180 kg/m <sup>3</sup> | $r_i$  | 0.045m |
| $E_{epoxy}$    | 4.35GPa                | $L_A$  | 0.5 m  |
| $G_{epoxy}$    | 1.59GPa                | $L_B$  | 0.5 m  |
| $\rho_{steel}$ | 7780 kg/m <sup>3</sup> | $m_A$  | 3      |
| $E_{steel}$    | 210.6GPa               | $m_B$  | 3      |
| $G_{steel}$    | 81GPa                  |        |        |

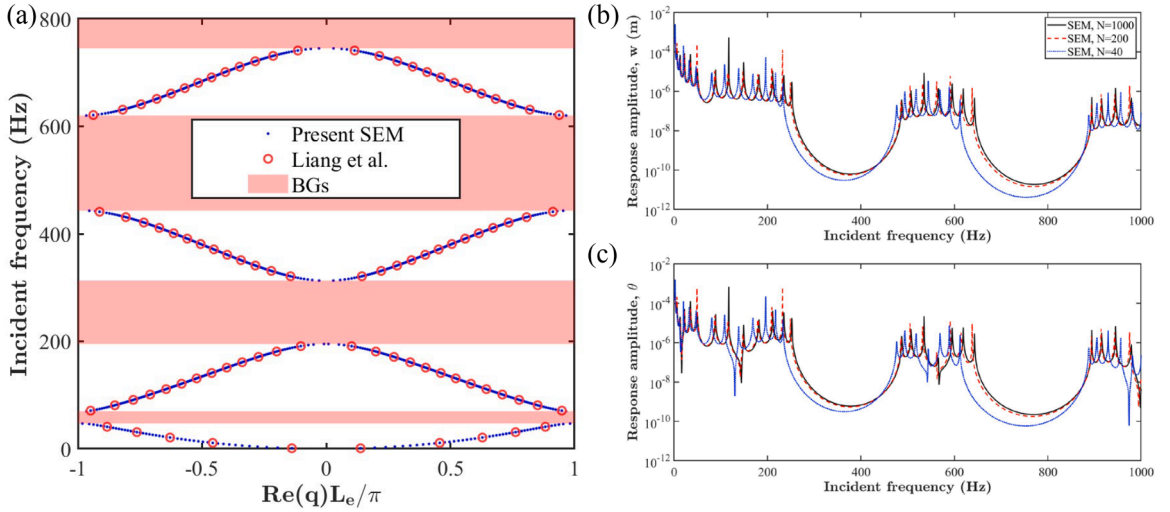


Fig. 4. Model verifications: (a) Model verification, (b, c) Frequency response convergence verification.

vector data presented in Fig. 4(a) are derived by adjusting the physical parameters of the current model to match the configuration of the periodical binary pipe conveying fluid described in Reference [44]. The detailed parameter variations are summarized as follows: Based on the default system parameters, the maximum outer radius  $r_o$  of the pipe is adjusted to 0.05 m, the power-law coefficient  $m_A$  and  $m_B$  are set to 0, and the material of section B of the pipe is specified as steel (see Table 1 for details). The results based on SEM demonstrate that the real part of the wave vector, obtained through parameter transformation of the metamaterial pipeline proposed in this study, is entirely consistent with the numerical analysis results of the binary PC pipeline reported by Liang et al., thereby validating the accuracy of the mechanical modeling approach employed in this work. Fig. 4(b) presents the frequency response of a metamaterial-inspired pipeline with preset parameters and a length of seven cells. In this case, a harmonic excitation of  $F = 1e^{i\omega t}$  N is applied at the left end of the pipeline, and the corresponding response signal at the right end is displayed. By comparing the frequency response results of the ABH unit divided into 40, 200, and 1000 elements, it can be observed that both the deflection and rotation frequency responses converge to a stable curve as the number of discretized units increases. This convergence behavior demonstrates the numerical stability of the discretization method employed for the ABH pipe in this study. Furthermore, the frequency response curves obtained with 200 and 1000 elements are nearly identical, indicating that the results have effectively converged. In contrast, using only 40 elements leads to a phase shift toward lower frequencies. Therefore, to ensure accuracy and reliability, both the SEM and

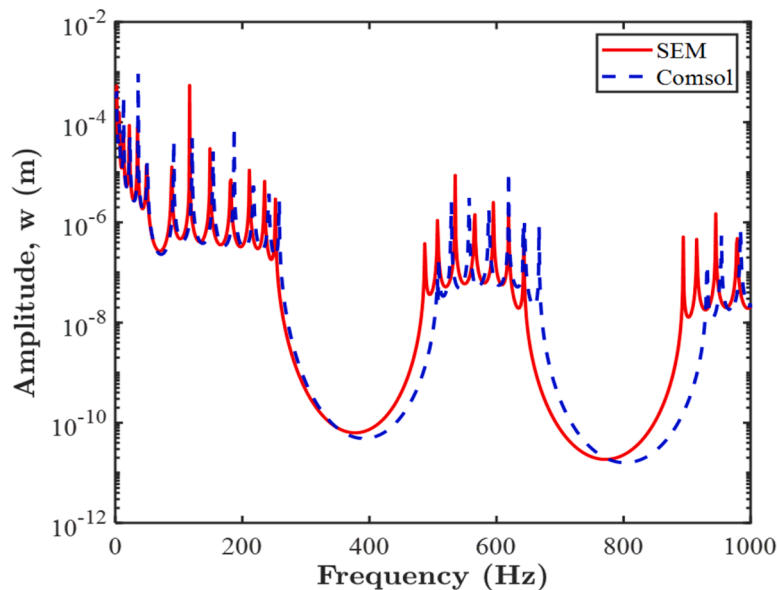


Fig. 5. Comparison of frequency response calculations with results from COMSOL Multiphysics.

TMM calculations in this paper are conducted using a discretization of 1000 elements.

Fig. 5 presents the frequency response validation results for the pipeline meta structure with a dual-material design. The red curve represents the solution computed using the SEM employed in this study, while the blue curve is obtained from finite element simulations performed in COMSOL Multiphysics (version 6.4). A total of 1,000 spectral elements are used in the SEM model, whereas the finite element model employs 26,436 tetrahedral elements. The two frequency response curves exhibit excellent agreement, particularly in the low-frequency range, where they are nearly identical, thereby validating the accuracy of the numerical approach adopted in this work. Small discrepancies observed at higher frequencies can be attributed to differences in spatial discretization and associated numerical dispersion errors inherent to each method.

Fig. 6 further validates the semi-analytical method employed in this study by comparing the results with experimental data. To facilitate experimental testing, a 10:1 geometrically scaled-down aluminum alloy specimen of a six-period pipeline meta structure is used in the validation process, with material and geometric parameters specified as follow:  $\rho_{Al} = 2800\text{kg/m}^3$ ,  $E_{Al} = 68.9\text{GPa}$ ,  $G_{Al} = 25.9\text{GPa}$ ,  $r_i = 4.4\text{mm}$ ,  $r_c = 5.7\text{mm}$ ,  $r_o = 10.3\text{mm}$  and  $L_A = L_B = 50\text{mm}$ . The frequency response characteristics are measured using a laser vibrometer system under a sweep frequency excitation scheme. The pipeline is fixed in a single-end clamped configuration, and a constant power harmonic excitation is applied. To acquire high-fidelity frequency response data, two strategically positioned response measurement points are implemented on the pipeline meta structure: one at the excitation application location (collection point 1, 100 mm from the clamped end), and the other (collection point 2) 100 mm from the free end. The complete experimental configuration is displayed in Fig. 6(a - d). Fig. 6(e) shows the velocity transmission from the collection point 1 to collection point 2, which is defined as  $20\log(\dot{w}_2/\dot{w}_1)$ . Based on Eq. (2), under the condition of constant excitation power, the vibration velocity of the response point in the SEM can be expressed as

$$\dot{w}_{1,2} = 2\pi i w_{1,2}. \quad (33)$$

A comparison between the simulated results obtained using the SEM and the experimental measurements over the 0–5000 Hz frequency range demonstrates that the SEM achieves high accuracy: it precisely captures the first seven resonance peaks of the scaled model, and the simulated velocity transmission curve exhibits excellent agreement with the experimental curve across the entire bandwidth. It should be noted that the discrepancy between the semi-analytical solution and the experimental measurements increases progressively with frequency. This trend can be attributed to two primary factors: (1) The finite stiffness of the experimental constraint,

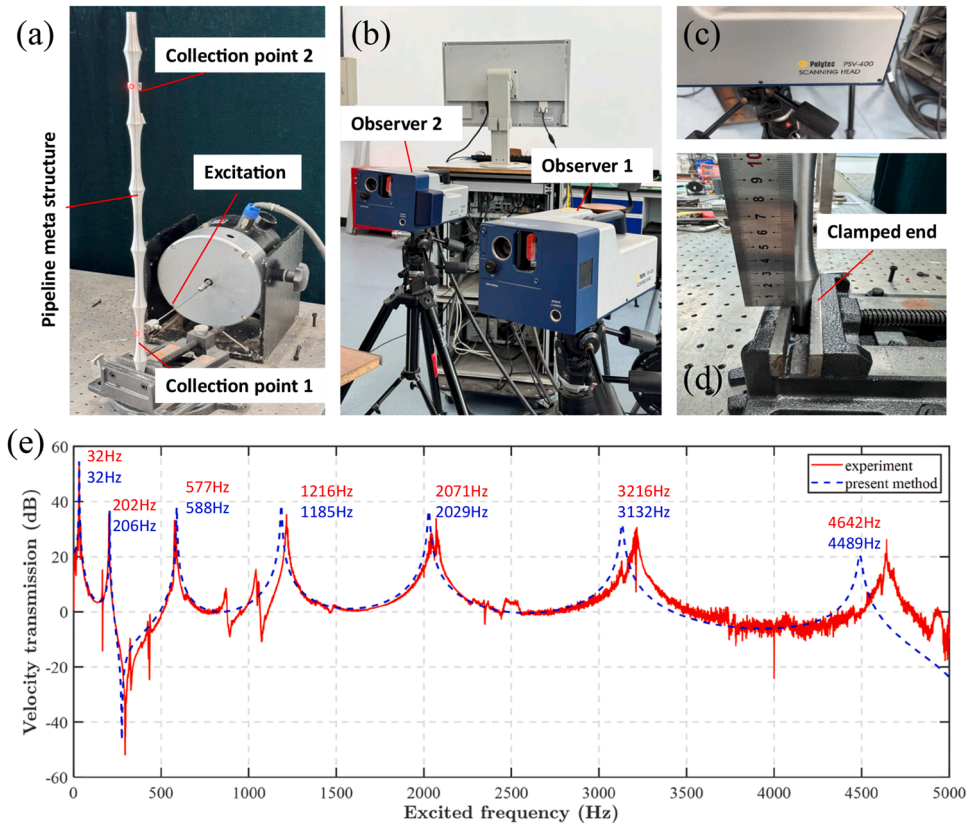


Fig. 6. Experimental verification: (a, b, c, d) Experimental platform, (e) Comparison between the semi-analytical and experimental results.

which is not accounted for in the semi-analytical model; (2) The geometric discretization inherent to the SEM employed in this study. At high frequencies, localized deformations become more pronounced, leading to an amplification of the truncation error associated with the discretization process.

#### 4.2. Band structures

The band structure of the metamaterial-inspired pipeline conveying fluid, composed of epoxy, is shown in Fig. 7. The real and imaginary parts of the wave vector reveal the presence of four BGs below 1000 Hz, specifically within the frequency ranges of 51 Hz – 66 Hz, 215 Hz – 272 Hz, 462 Hz – 516 Hz, and 740 Hz – 780 Hz. Fig. 7(c) further presents the frequency response function (FRF) at the right end of a 7-cell metamaterial-inspired pipeline with no constraints, under a periodic excitation force of  $F = 1e^{i\omega t}$  N applied at the left end. The FRF is defined as

$$\text{FRF} = 20\log(w_{\text{response}} / w_{\text{excitation}}). \quad (34)$$

The FRF result indicates that within the BG predicted by the wave vector analysis, the pipeline effectively attenuates the vibration induced by the input excitation (as evidenced by the negative FRF values). This demonstrates that the BG of the metamaterial pipeline corresponds precisely to the frequency range where structural stability and vibration suppression are achieved. Fig. 8 presents the band structure of the metamaterial-inspired pipeline conveying fluid based on the dual-material design. The geometric configuration remains consistent with that illustrated in Fig. 2, where the part-A is composed of epoxy and the B-part is made of steel. The dual-material design not only induces structural impedance variations through the ABH structure but also generates additional impedance discontinuities via periodic material alternation. Below 1000 Hz, three distinct BGs of 52 Hz – 75 Hz, 246 Hz – 480 Hz and 630 Hz – 886 Hz are observed. In comparison with the single-material design, the second and third BGs in this configuration exhibit significant widening. Furthermore, the larger values of the imaginary part of the wave vector within the BG suggest a pronounced vibration attenuation capability. This prediction is substantiated by the results of the FRF calculations. In fact, in the design of PC pipes based on Bragg scattering, a wide BG not only signifies an expanded frequency range for vibration reduction, but also reflects enhanced vibration attenuation performance within that range.

Fig. 9 displays the frequency response of a 4-cell metamaterial-inspired pipeline composed of epoxy. In this example, a harmonic excitation  $F = 1e^{i\omega t}$  is applied at a location that is one-quarter of the total length from the left end of the pipe, while the response signal at a corresponding position one-quarter of the total length from the right end is recorded and plotted. The results demonstrate that the pipeline exhibits varying degrees of response attenuation within the band gap predicted by the wave vector under three different boundary conditions: free (unconstrained), simply supported at both ends, and fixed at both ends. For the constrained cases (pinned-pinned and fixed-fixed), the amplitude attenuation within the BG is more pronounced. In contrast, under free boundary conditions, the frequencies exhibiting the maximum amplitude attenuation are located in the frequency range adjacent to the BG. This behavior is likely attributed to the influence of the pipeline's rigid body modes. When the number of pipe periods is sufficiently large, the pipeline can be treated as a flexible structure, and the influence of rigid body modes on the response amplitude becomes negligible. In contrast, for a small number of periods, rigid body displacement dominates the dynamic response, leading to a poor correspondence between the minimum amplitude and the BG. It is noted that the amplitude attenuation shown in Fig. 9 is only caused by two pipe cells, and the vibration reduction effect of the BG seems not significant. To demonstrate the effect of the number of periods on the wave retardation, Fig. 10 further presents the frequency response of the pipe without constraint under different numbers of cells. Fig. 10 employs the same loading and response acquisition methods as those used in Fig. 9, with the number of pipe cells configured as 4, 8, and 12, respectively. It is observed that the amplitude response within the first four BGs gradually decreases as the number of pipe units increases. The amplitude reduction effect of the BGs is consistently manifested in both the deflection response and the bending angle response. The native BG frequency range of the single-material pipeline meta structure is relatively narrow, and a single unit cell exhibits limited effectiveness in suppressing bending wave propagation. Consequently, stacking multiple pipeline periods is necessary to achieve a more pronounced vibration reduction performance.

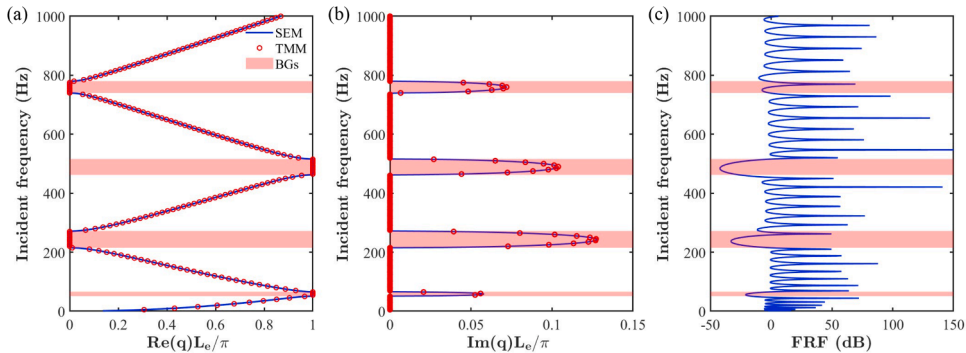
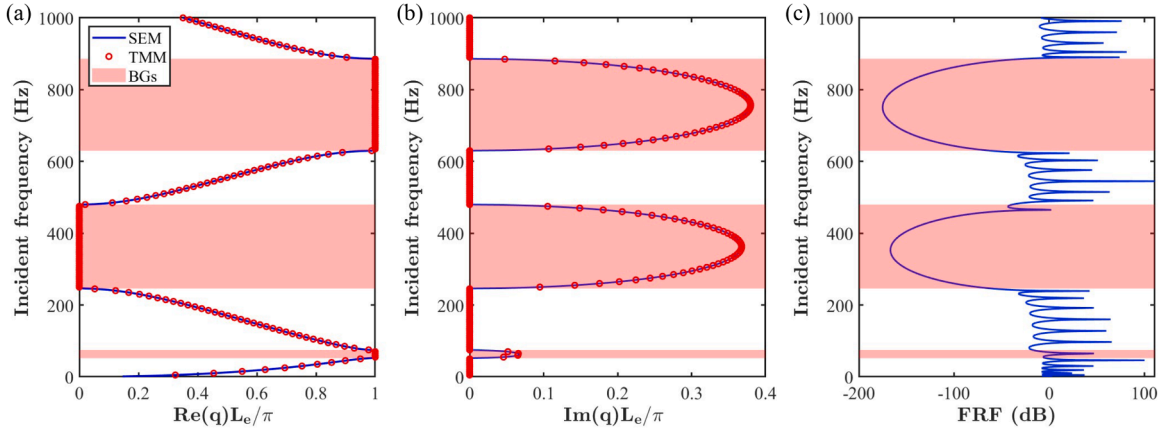
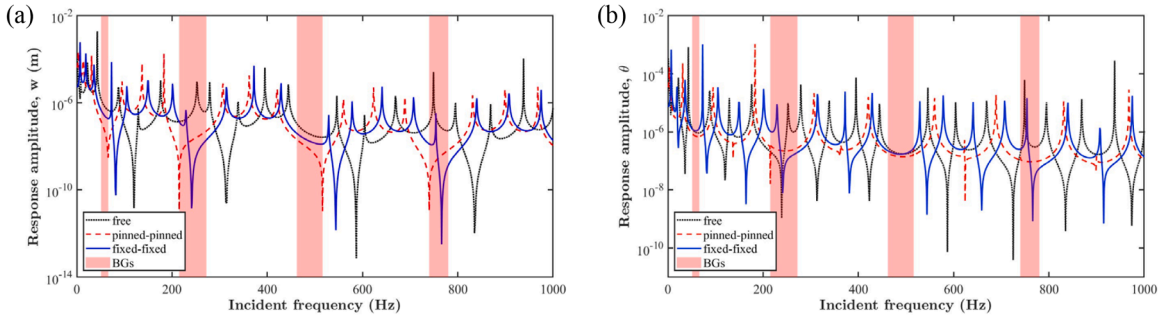


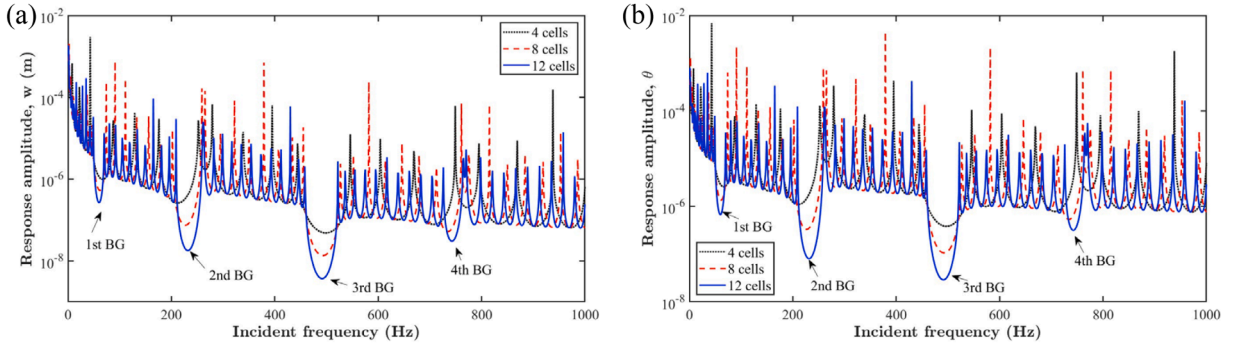
Fig. 7. Band structure of the single-material design: (a) Real part of the wave vector, (b) Imaginary part of the wave vector, (c) FRF of the 7-cell pipeline conveying fluid.



**Fig. 8.** Band structure of the dual-material design: (a) Real part of the wave vector, (b) Imaginary part of the wave vector, (c) FRF of the 7-cell pipeline conveying fluid.



**Fig. 9.** Frequency response of the single-material design: (a)  $w$ , (b)  $\theta$ .



**Fig. 10.** Frequency response of the metamaterial-inspired pipeline with different period number: (a)  $w$ , (b)  $\theta$ .

Fig. 11 presents the frequency response characteristics of the dual-material designed pipeline meta structure under various constraint scenarios. The loading configuration and response measurement approach employed in this case are identical to those used in the example shown in Fig. 9. Compared to the single-material design, the dual-material design of the pipeline meta structure demonstrates a substantially reduced response amplitude within the wave vector-predicted BG under all three constraint scenarios, encompassing both deflection amplitude and angle amplitude. It is worth noting that the second and third BGs are classified as wide BGs. Under different constraint conditions, the amplitude responses within these wide BGs are markedly attenuated. Furthermore, the influence of constraints on the amplitude responses within these wide BGs remains minimal.

Fig. 12 presents the energy distribution within a single unit cell of the two proposed pipeline meta structures under external excitation. A displacement excitation of  $w = 0.001e^{i\omega t}$  m is applied at one end of the pipeline to simulate forced vibration conditions. The kinetic energy and strain energy of the segment of  $[x_1, x_2]$  along the pipeline can be expressed as:

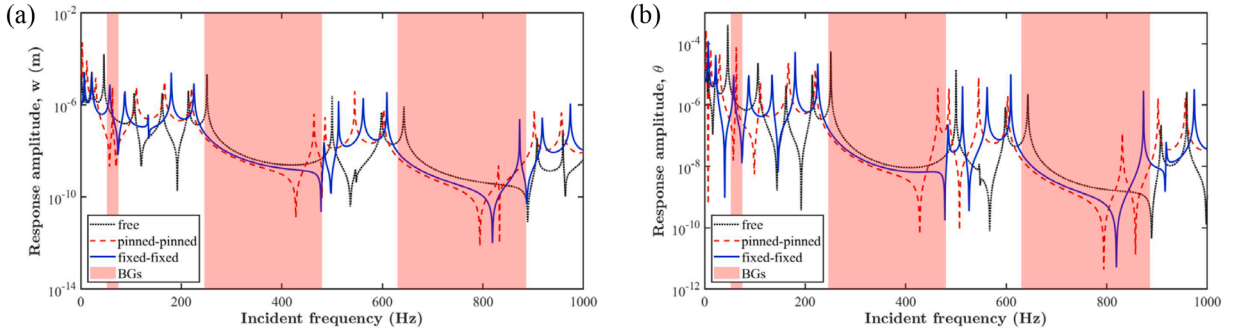


Fig. 11. Frequency response of the dual-material design: (a)  $w$ , (b)  $\theta$ .

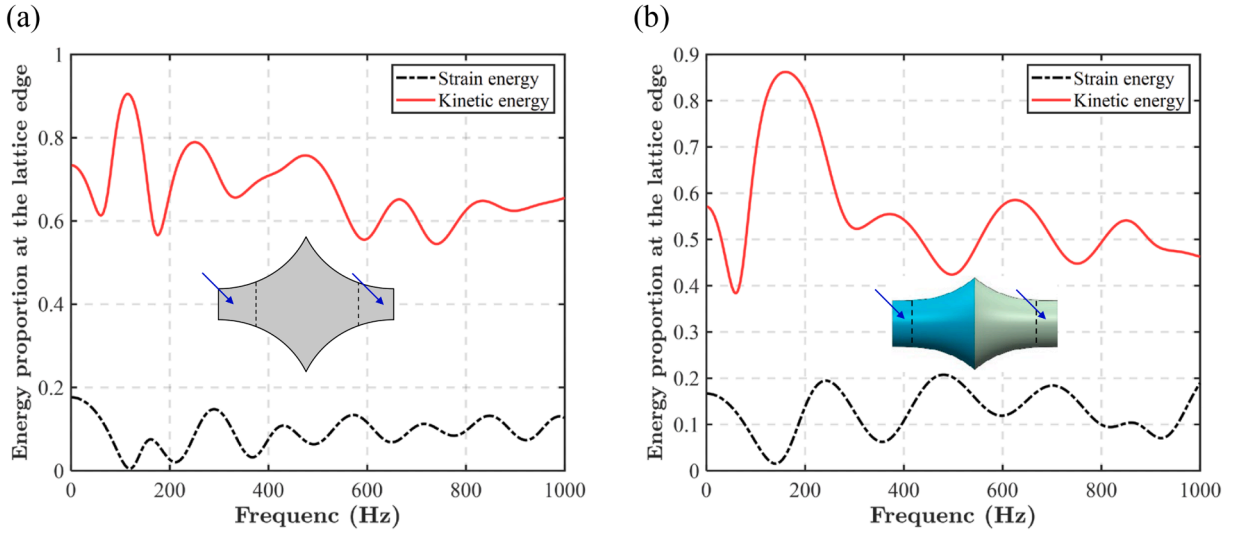


Fig. 12. Energy distribution in the lattice, (a) Single material, (b) Dual material.

$$\begin{cases} V_s(x_1, x_2) = \int_{x_1}^{x_2} EI \left( \frac{\partial \theta}{\partial x} \right)^2 + \kappa GA \left( \frac{\partial w}{\partial x} - \theta \right)^2 dx \\ V_k(x_1, x_2) = \int_{x_1}^{x_2} \frac{1}{2} (m_f + m_p) \left( \frac{\partial w}{\partial t} \right)^2 dx \end{cases}, \quad (35)$$

where  $V_s$  and  $V_k$  represent the strain energy and kinetic energy, respectively. According to the ABH theory, elastic waves tend to localize at the edge regions of the lattice. To quantify this effect, Fig. 12 presents the energy distribution ratios over the leftmost and rightmost 1/10 segments of the unit cell for both design configurations. The results demonstrate that, in both single-material and dual-material designs, the ABH effect is effectively activated within the lattice structure. Specifically, strain energy at the edges remains relatively low, whereas kinetic energy is significantly concentrated near the boundaries. Across all frequency bands, the edge kinetic energy accounts for more than 55% of the total kinetic energy in the single-material design and exceeds approximately 40% in the dual-material design, despite the edge mass constituting an extremely small fraction of the total. This pronounced energy localization confirms that the ABH effect synergistically enhances Bragg scattering, thereby representing a key advantage of the proposed structures.

Under the initial parameter settings, the BG width achieved by the single-material pipeline meta structure is narrower compared to that of the dual-material design, resulting in relatively weaker vibration attenuation performance within the BG. The dual-material configuration enhances vibration suppression capabilities by combining material periodicity with geometric periodicity. However, this approach also increases the overall pipe mass and manufacturing complexity. In contrast, the periodically arranged ABH fluid-conveying pipeline proposed in the authors' prior work, based on a single-material and unidirectional ABH structural design, exhibits a significantly broader BG than the single-material design presented in this study. This suggests that, under the default parameter settings, the pipeline meta structure introduced in this paper does not fully capitalize on the BG-expanding potential of the ABH

structure. Strategies for enhancing the BG performance of the proposed pipeline meta structure through systematic parameter optimization will be explored in the following sub-sections.

#### 4.3. FSI characteristics

The dynamic properties of the pipeline conveying fluid, such as its stability and band structure, are influenced by the inertial force, Coriolis force, and centrifugal force generated by the conveyed fluid, and these characteristics vary with changes in the flow velocity. Fig. 13 displays the variation in the first-order natural frequency of the metamaterial-inspired pipeline as a function of flow velocity. For the single-material pipeline meta structure, when the lattice length is kept constant and the lengths of parts -A and -B are varied, the relationship between the natural frequency and flow velocity exhibits only minor variations. Fig. 13(a) presents the first-order natural frequencies of Part A and Part B with length ratios of 1/3, 1/2, and 1 under the single-material design. The three curves are closely aligned. As the flow velocity increases, the first-order natural frequency (real part) diminishes, indicating the onset of pitchfork instability in the pipeline. It can be observed that the asymmetric lattice structure lowers the critical flow velocity, thereby enhancing the stability of the pipeline meta structure. The dynamic stability of the pipeline meta structure is markedly influenced by the implementation of a dual-material design. As illustrated in Fig. 13(b), the first natural frequency of the pipeline varies with flow velocity under this design configuration, where Part -A is composed of epoxy and Part -B is constructed from steel. Due to the substantial impedance mismatch between parts -A and -B, modifying the length ratio of these segments significantly influences the first-order natural frequency, thereby ultimately altering the critical flow velocity of the pipeline meta structure. The results indicate that, under identical flow conditions, designs featuring a higher proportion of Part -B (steel) exhibit an increased first-order natural frequency, which consequently enhances the dynamic stability of the system.

Fig. 14 shows the effect of flow velocity on the BGs of the pipeline meta structure. In particular, Fig. 14(a) depicts how the BGs of the single-material configuration varies with flow velocity under default parameter settings, whereas Fig. 14(b) demonstrates the corresponding behavior for the dual-material design. In this case, both pipeline meta structures are configured as four cells in length and feature simply supported boundary conditions at both ends. The results indicate that flow velocity exerts a minor influence on the BG characteristics of both pipeline meta structures. With increasing flow velocity, the BGs of both configurations exhibit a slight downward shift toward the low-frequency range. Overall, increasing the flow rate of the transported liquid facilitates the low-frequency shift of the band gap. However, once the critical flow velocity is exceeded, the pipeline meta structure becomes unstable, which compromises its effectiveness in vibration control. Furthermore, due to the use of a linear motion equation in this study, the system may exhibit nonlinear vibrational behavior under supercritical flow conditions, leading to increased errors in the calculated band gaps beyond the critical flow velocity.

#### 4.4. Parameter covariance characteristic

The proposed pipe meta structure generates BGs through a periodic ABH lattice arrangement, and the design parameters of the lattice directly determine the distribution and width of these BGs. Given that the ABH lattice involves multiple covariant parameters, optimizing their settings to broaden the BGs is of considerable engineering significance, as it enhances the vibration suppression performance of the pipeline system.

Fig. 15 displays how the BG characteristics of the pipeline meta structure vary with the power-law coefficient at the ABH lattice

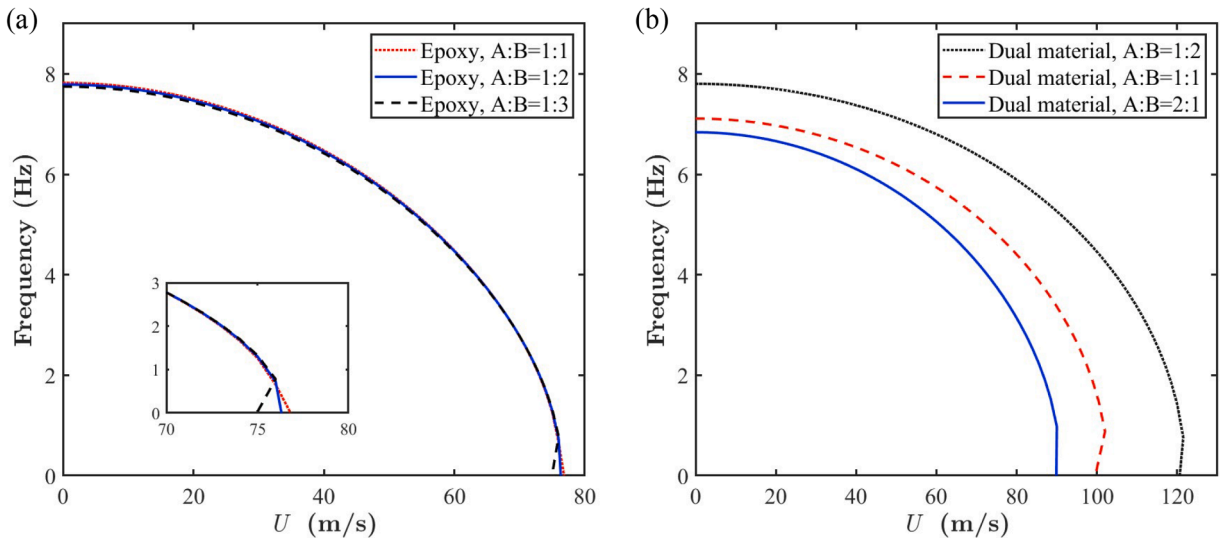


Fig. 13. The first-order natural frequency varying with fluid velocity: (a) single-material design, (b) dual-material design.

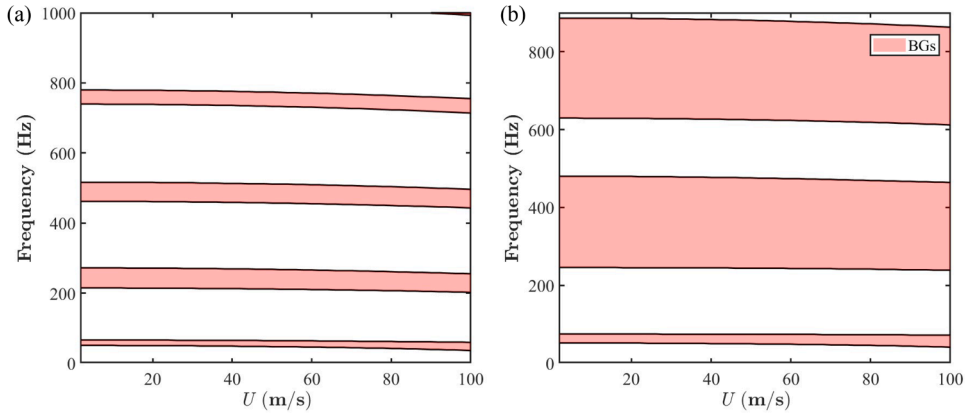


Fig. 14. BGs varying with fluid velocity: (a) single-material design, (b) dual-material design.

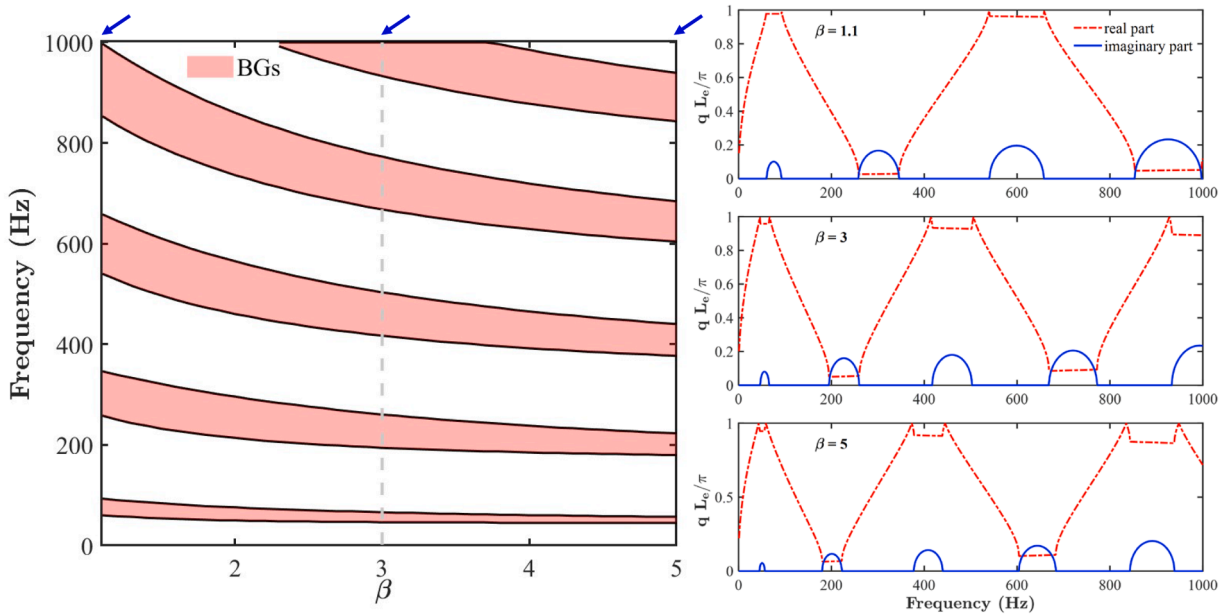


Fig. 15. BGs are presented as a function of the power-law exponent of the ABH lattice, single-material design.

edge under a single-material design. The results demonstrate that as the power-law coefficient  $\beta$  increases, all BGs shift toward lower frequencies, and their corresponding bandwidths gradually narrow. Moreover, the rate at which the bandgap shifts toward lower frequencies gradually decreases with increasing  $\beta$ . The results indicate that when  $\beta$  exceeds 5, the low-frequency bandgap ceases to shift significantly, approaching a limiting frequency. This trend introduces two competing effects on the vibration suppression performance of the pipeline meta structure, thereby making it challenging to achieve both low-frequency and broadband vibration attenuation simultaneously through adjustments to the value of  $\beta$  alone. It is noteworthy that as  $\beta$  increases, a plateau emerges in the real part of  $q \times \frac{L_0}{\pi}$  within the attenuation band (the frequency range where  $\text{Im}(q) \times \frac{L_0}{\pi} \neq 0$ ), gradually deviating from the values of 0 and 1. This observation suggests that certain levels of vibration energy remain transmissible within the band gap, thereby reducing the effectiveness of vibration damping in that region. This phenomenon may also account for the anomalous fluctuations observed in the frequency response curve within the band gap under default parameter settings, as shown in Fig. 9.

Fig. 16 shows the variation of BGs as  $r_o$  increases. For clarity, the horizontal axis in the left panel is expressed in the dimensionless form of  $r_o/r_c$ . The results demonstrate that for the designed pipeline meta structure, the maximum outer diameter of the ABH structure is a key parameter for tuning the BG. By increasing this geometric dimension, the BG range can be significantly broadened, enabling effective wideband vibration attenuation. When  $r_o/r_c$  exceeds 5, the first four BGs collectively cover 82% of the frequency range below 1000 Hz. More importantly, as  $r_o/r_c$  increases, the real part of  $q \times \frac{L_0}{\pi}$  within the attenuation band progressively approaches exactly 0 or 1, indicating the formation of true stop bands rather than mere wave attenuation. Nevertheless, expanding the outer diameter to improve vibration suppression performance entails trade-offs, specifically, an increase in both the physical dimensions and mass of the

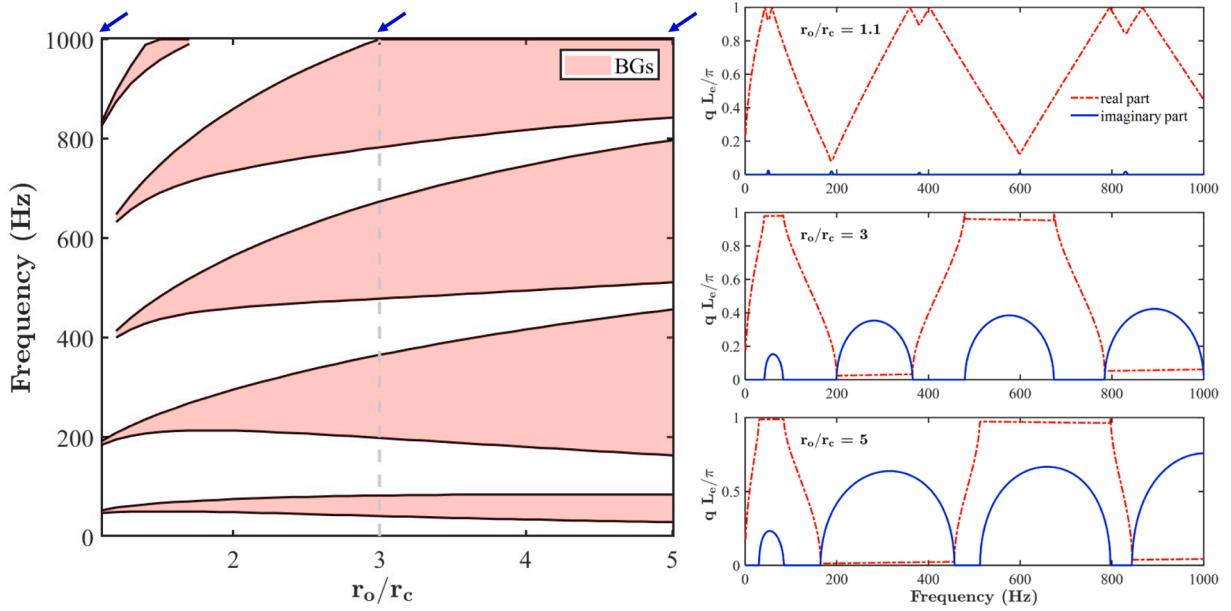


Fig. 16. BGs are presented as a function of the ratio between the maximum and minimum outer diameters of the ABH lattice, single-material design.

pipeline meta structure, which may limit its applicability in certain engineering contexts.

In the proposed pipeline meta structure, each cell incorporates two ABH sub-units arranged in opposite orientations. This hierarchical configuration enables more sophisticated dynamic wave manipulation and directional design. Fig. 17 shows the influence of the length distribution between the two ABH sub-units on the BG characteristics. In our calculations, the total length of a single lattice is fixed at 1 m. Under the original parameters,  $L_A = L_B = 0.5$  m, resulting in a symmetric lattice in the single-material design. It is observed that this symmetric configuration yields the narrowest BG in the plot. Deviating from this symmetry, either increasing or decreasing  $L_A$ , leads to a significant broadening of the BGs, particularly for the third and fourth BGs. The left panel of Fig. 17 is symmetric about  $L_A = 0.5$  m, indicating that increasing the length proportion of either part-A or part-B yields equivalent effects on the BG. This symmetry is further confirmed by the wave vector analysis in the right panel of Fig. 17. This case demonstrates that geometric asymmetry within a single lattice promotes the formation of broad BGs in the pipeline meta structure.

In this work, an alternative feasible design of the pipeline meta structure is proposed by employing distinct materials for part-A and part -B, while ensuring that the subunits still conform to the ABH geometric configuration, thus enabling the formation of a lattice structure composed of dual-material ABH assemblies. Fig. 18 displays the influence of the ABH power-law exponent on the BGs in the context of the dual-material design. The results demonstrate that with increasing ABH power-law exponent, all BGs shift progressively toward lower frequencies, a trend that closely resembles the behavior observed in single-material designs. Notably, in the dual-material configuration, the power-law exponent  $\beta$  exerts minimal effect on the widths of the first and second BGs, however, for higher-frequency BGs (such as the third BG), the width gradually decreases as the exponent increases. This characteristic allows the dual-material design to effectively enhance low-frequency vibration attenuation through an appropriate increase in the power-law exponent. As revealed by the wave vector diagram, the BGs in the dual-material design remain the strict stop bands across all values of the power-law exponent. In addition, it is worth noting that when  $\beta$  equals zero, the dual-material pipeline meta structure proposed in this work reduces to a conventional binary PC pipe. Under comparable geometric dimensions, the binary PC pipe exhibits a higher bandgap frequency range compared to the dual-material meta structure, highlighting the effectiveness of the ABH configuration in enabling superior low-frequency vibration attenuation performance.

In the dual-material design, the BG distribution of the pipeline meta structure can be effectively tuned by adjusting the maximum outer diameter of the ABH sublattice, as illustrated in Fig. 19. Clearly, increasing the maximum outer diameter of the ABH sub-lattice can further enhance the ABH effect and significantly broaden the BGs. This trend is consistent with the behavior observed in single-material designs, as illustrated in Fig. 16. However, in the dual-material design, a periodic arrangement formed by alternating materials emerges when the maximum outer diameter equals the minimum inner diameter. This inherent periodicity ensures a substantially broader BG coverage across all values of the maximum outer diameter parameter, highlighting a distinct advantage over single-material configurations. Moreover, across all values of  $f_o/r_c$ , the real part of  $q \times L_c/\pi$  within the band gap is strictly either 0 or 1, indicating that the BG in the dual-material design remains a stop band irrespective of the strength of the ABH effect. The dual-material design enables the achievement of an ultra-wide BG through a slight increase in the value of  $f_o/r_c$ , facilitated by the synergistic interaction between periodic material distribution and periodic geometric configuration.

Fig. 20 shows the influence of the length ratio between sublattices part-A and part-B on the BG of the pipeline meta structure in the dual-material design. For clarity, in the left panel of Fig. 20, the horizontal axis represents the length  $L_A$  of part-A (epoxy), with the

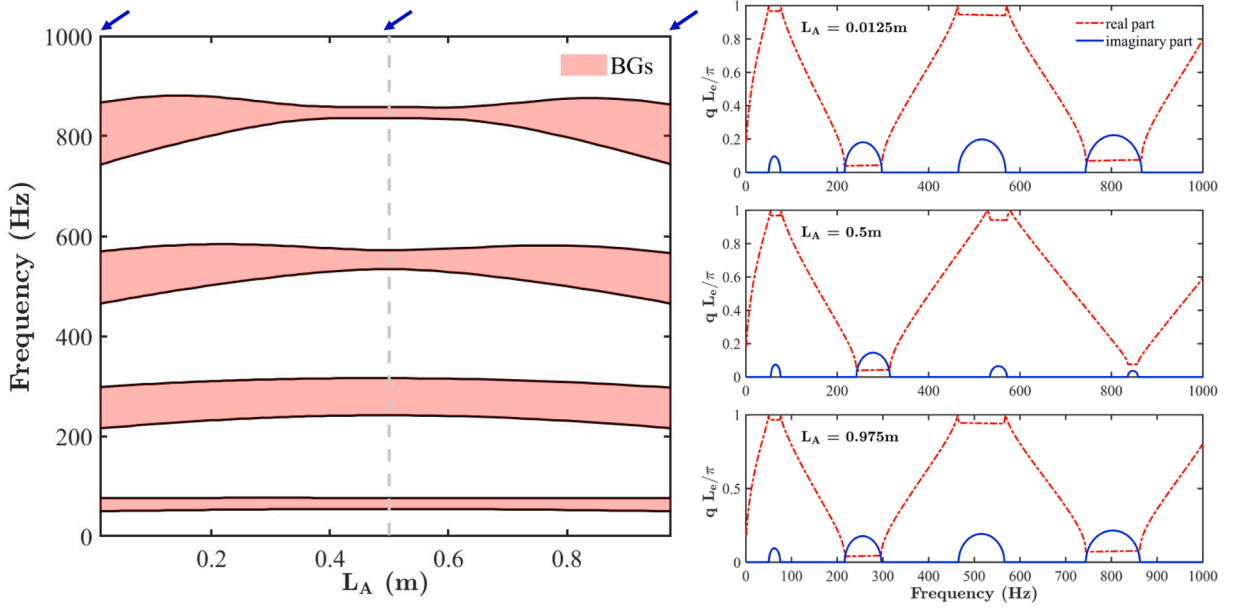


Fig. 17. BGs are presented as a function of the length proportion of the sub-lattices.

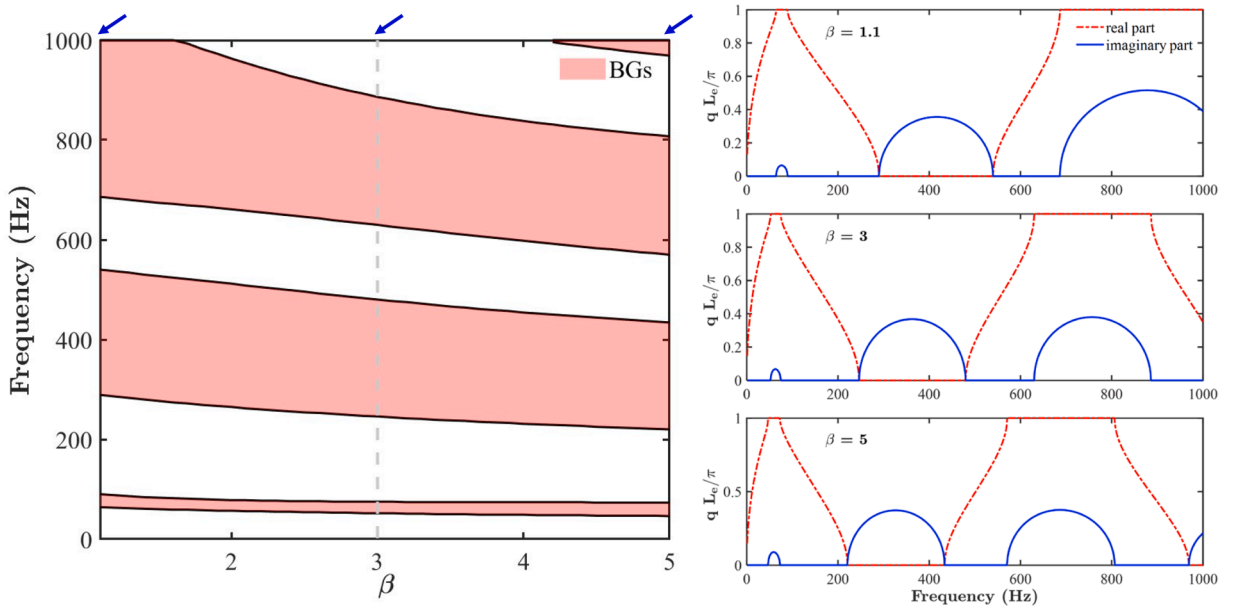


Fig. 18. BGs are presented as a function of the power-law exponent of the ABH lattice, dual-material design.

length of part-B (steel) defined as  $L_B = 1 - L_A$ . The results indicate that an increased length of the iron-based part-B section leads to a broader native BG in the pipeline meta structure. This behavior arises from the higher elastic modulus and density of steel, which collectively result in elevated impedance of part-B. However, as the length of the epoxy sublattice increases, the overall mass of the pipeline meta structure decreases, while the BG exhibits a more complex evolution. Specifically, the onset frequencies of the BGs shift toward lower frequencies, and the bandwidths first increase and then decrease. Analysis of the wavevector computation results reveals that the design with steel as the dominant lattice component (characterized by  $L_A = 0.04$  m) yields a BG with a higher imaginary part of  $q \times \frac{L_0}{\pi}$ , indicating a significant attenuation of structural response amplitude within the BGs, albeit at the expense of increased overall structural mass. In contrast, configurations where epoxy occupies the majority of the lattice length exhibit narrower BGs, thereby limiting their practical vibration suppression capability. These findings demonstrate that the dual-material sublattice design plays a crucial role in enhancing the low-frequency broadband vibration isolation performance of the pipeline meta structure. For the

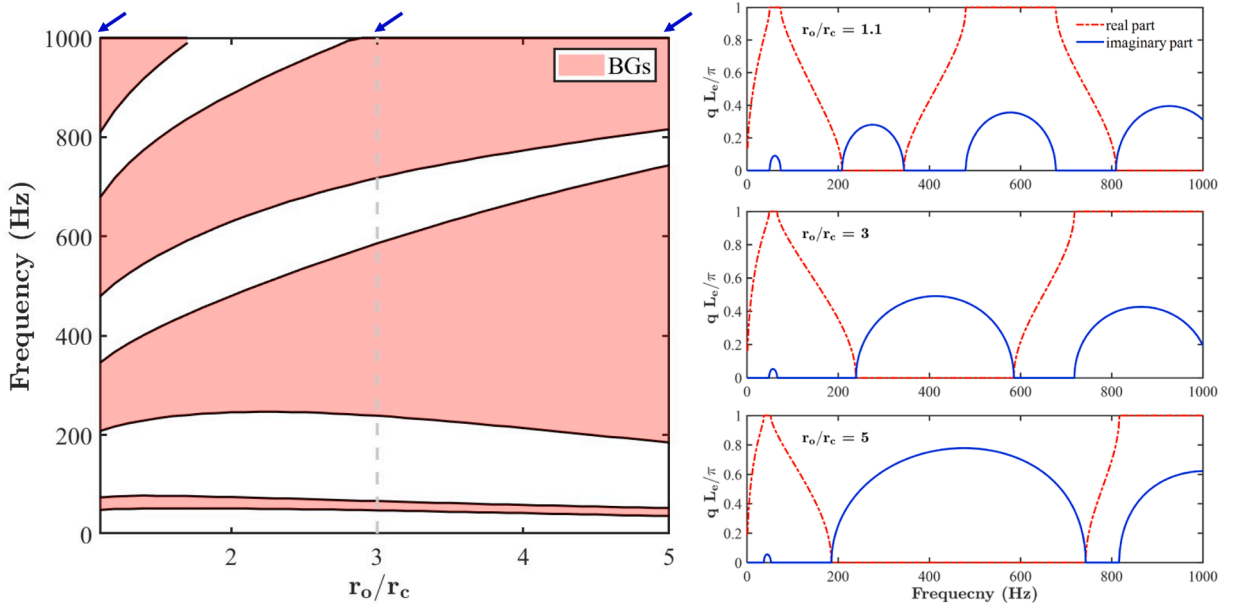


Fig. 19. BGs are presented as a function of the ratio between the maximum and minimum outer diameters of the ABH lattice, dual-material design.

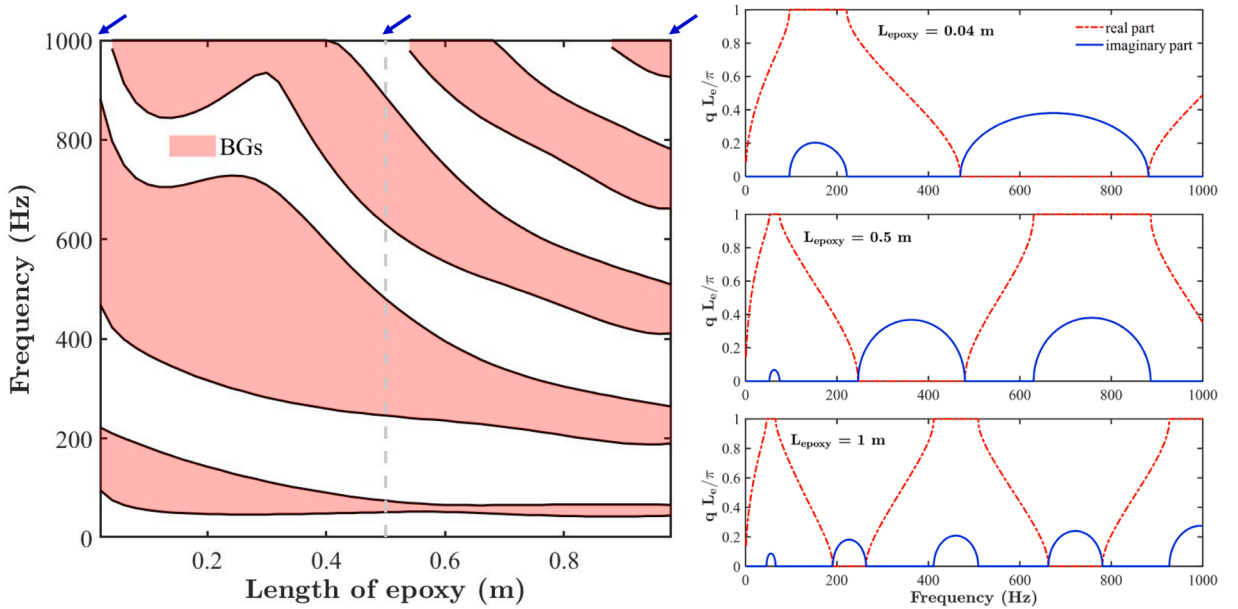


Fig. 20. BGs are presented as a function of the length ratio between the two materials.

proposed structure in this study, an optimal material distribution is achieved at  $L_A = 0.25$  m.

For the proposed pipe meta structure, two dominant factors critically influence its band structure characteristics: the sublattice distribution and the ABH geometric parameters. By tuning these design parameters, the BGs can be effectively shifted, widened, or narrowed, thereby enabling precise control over BG properties and establishing a foundation for directional BG design. The dual-material design exhibits a wider BG than its single-material counterpart under identical geometric parameters. However, this does not imply universal superiority. The increased mass and greater manufacturing complexity associated with the dual-material configuration constrain its practical applicability in engineering contexts. The pipeline meta structure proposed in this work enables a wider BG more readily than the conventional binary PC pipes [19]. This advantage stems from the reverse ABH geometric configuration, which constitutes a quasiperiodic design. Recent studies have demonstrated that in metamaterial design, quasiperiodic structures are more effective than periodic ones in achieving low-frequency and broadband BGs [27,45].

A systematic analysis of the influence of system parameters on BG's formation reveals that symmetric sub-lattice configurations are highly detrimental to BG generation. Therefore, in single-material designs, geometric symmetry should be minimized. Effective strategies include avoiding equal lengths of part-A and part-B and employing distinct power-law coefficients for the two sub-lattices. In contrast, dual-material designs inherently possess asymmetry due to differing material properties between sub-lattices, resulting in substantial band gaps even under geometrically symmetric configurations. A notable special case of asymmetric design occurs when the length of part-B is zero, with only part-A retained, under this condition, the proposed pipeline meta structure reduces to the periodic ABH pipes previously proposed by the authors [42]. While such unidirectional ABH lattice arrangements facilitate the realization of broadband BGs, the geometric discontinuities at lattice junctions act as inherent geometric defects, potentially leading to stress concentration or structural fatigue. Whether through material distribution or geometric configuration asymmetry, the primary objective is to establish impedance asymmetry within the lattice structure. In ABH configurations, a high impedance gradient significantly enhances the bandgap width of the periodic system. For example, as shown in reference [35], a periodic ABH beam employing functionally graded materials achieves such a steep impedance gradient, resulting in a substantially broader bandgap.

It is also worth noting that for both single-material and dual-material designed pipeline meta structures, the first bandgap is narrower than the second and third bandgaps, resulting in relatively limited amplitude attenuation in short pipelines with fewer unit cells. However, this does not indicate poor low-frequency vibration suppression performance of the proposed structure. As revealed by the frequency response results, the extent of amplitude attenuation within the first bandgap is highly dependent on the number of periodic units. When the number of cycles is sufficiently large, significant attenuation can be achieved even within the first bandgap. This behavior constitutes a key advantage of the proposed meta structure for low-frequency vibration control in pipeline systems. Specifically, short pipelines typically exhibit higher fundamental natural frequencies, where the need for low-frequency vibration mitigation is minimal. In contrast, long pipelines (commonly encountered in engineering practice) possess lower fundamental frequencies, making effective suppression of low-frequency vibrations critically important. By implementing a multi-cycle meta structure design in such long pipelines, substantial reduction in vibrational amplitude within the bandgap can be achieved. The inherent property that bandgap-induced attenuation increases with the number of cycles enables a highly flexible design approach: through appropriate tuning of geometric and material parameters to align the bandgap with the target frequency range, the proposed structure can effectively satisfy the vibration reduction requirements across a broad spectrum of practical pipeline applications.

## 5. Conclusion

In this work, a pipeline meta structure for low-frequency and broadband vibration suppression is proposed through the integration of a sublattice structure into a periodic PC pipeline. The sublattice is implemented via a pair of anti-parallel ABH pipelines, which introduces inherent asymmetry and enables tunable covariant BG behavior governed by geometric parameters. Furthermore, by altering the material of one sublattice in the dual-sublattice configuration, the system achieves enhanced vibration suppression performance due to increased structural and impedance mismatch. The innovations and findings of this work are summarized as follows:

1. For the proposed pipeline meta structure, bandgap width constitutes the most critical vibration damping performance metric. This significance arises not only from its role as the effective frequency range for wave attenuation, but also from the strong correlation between bandgap width and structural response within the bandgap. Specifically, a wider bandgap is associated with lower response amplitudes, thereby yielding superior vibration suppression performance.
2. The flow velocity of the conveyed fluid primarily influences the pipeline's stability. With increasing flow velocity, the BG gradually shifts toward lower frequencies. However, this effect remains negligible below the critical flow velocity.
3. For both single-material and dual-material designs, the geometric parameters of the sublattice ABH configuration serve as the primary means of bandgap tuning. Increasing the power-law coefficient of the ABH profile shifts the bandgap toward lower frequencies, whereas enlarging the maximum outer diameter of the ABH significantly widens each bandgap, thereby enhancing broadband vibration suppression performance.
4. The asymmetry between the two ABH sublattices positively influences bandgap expansion. This asymmetry arises either from differences in the geometric parameters of part-A and part-B, or from variations in their material compositions.

## CRedit authorship contribution statement

**Yang Bu:** Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ye Tang:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Dongyang Chen:** Validation, Supervision, Resources, Funding acquisition. **Tianzhi Yang:** Supervision, Resources, Methodology, Conceptualization. **Qian Ding:** Supervision, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

This work is supported by the National Natural Science Foundation of China (Nos. 12572025, 12132010, 52471342), Fundamental Research Funds for the Central Universities (No. D5000240076), Natural Science Basic Research Program of Shaanxi (No. 2025JC-YBMS-026), Beijing Natural Science Foundation (No. 3262011), Anhui Provincial Housing and Urban-Rural Development Science and Technology Plan Project (No. 2025-YF106), Innovation Team Project for Natural Sciences in Universities of Anhui Province (No. 2024AH010005), Wuhu Science and Technology Planning Project (No. 2024kj024), Open Fund for the Engineering Research Center of Anhui Green Building and Digital Construction (AHSGCZX202502) and Basic Research Programs of Taicang (No. TC2024JC36).

We would like to thank Tao Wang from Tianjin University for his assistance in the experiments.

## Data availability

Data will be made available on request.

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